

# WEB PROGRAMMING STUDIO

## ASSESSMENTS 1 - 3

Version 2.8-vn

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| Assessment Type                    | Group Project, Consisting of 3 Submittable Components                                                                                                                                                                                                                                                                                          |
| Due Dates                          | On the following dates, by 11:59 pm:<br><ol style="list-style-type: none"><li>1. Design Document: Friday of week 4</li><li>2. Prototype: Friday of week 8</li><li>3. Full-stack web application and<br/>Group presentations with a demo video: week 11</li></ol>                                                                               |
| Marks (percentage of total course) | <ul style="list-style-type: none"><li>• 20% - Design Document &amp; Static web pages</li><li>• 20% - Dynamic front-end web application</li><li>• 20% - Product presentation and Full stack web application</li><li>• (The remaining 40% of course marks come from the Full stack in-class Lab Test, detailed in a separate document)</li></ul> |

### A. INTRODUCTION

In Web Programming Studio, the assessment activities are interconnected to form a major group project. Working collaboratively in this group project, you will design and develop an interactive, fully functional web application that integrates both client-side and server-side components, operates toward a specific topic or theme of your choice. This project enables you to apply and extend the technical and design skills developed throughout the course.

This project comprises of three assessments in relation to three stages reflecting an authentic web development process — from planning and design through to implementation, testing, and presentation. Your team will decide on the theme or application focus of the project. Regardless of the theme, the final product must meet a set of technical and functional requirements provided in the assessment brief.

This assessment evaluates not only your technical proficiency in coding and design but also your collaboration, communication, and innovation skills. It measures both team performance and individual contribution, acknowledging that professional web development requires effective teamwork alongside personal expertise.

### B. TEAMWORK AND COLLABORATION GUIDELINES

## Teamwork and Project Structure

For this project, you will work in a team of four members, reflecting an authentic professional web development environment. The assignment requires you to collaborate productively and manage your project as a cohesive team.

## Collaboration and Conflict Resolution

Teams are expected to work collaboratively and resolve challenges through open communication. If disagreements or conflicts arise, the group should make every reasonable effort to reach a resolution independently. If a resolution cannot be achieved, you must inform a staff member and provide evidence of the steps your team has taken to address the issue. The staff member will then facilitate a resolution process.

## Module Responsibilities

*Each team member* will take the lead on one of five key modules:

1. Shopping Cart
2. Discussion Forum
3. Blog
4. Product Review and Rating
5. Wishlist (or favorite products)

Each module leader is responsible for demonstrating the full range of technical, functional, and professional competencies outlined in the marking rubric - including planning, design, implementation, client-side programming, and documentation.

In addition to these modules, the team will jointly develop a shared fifth module. All members must contribute equally to this shared component.

Although individual modules have distinct ownership, the final website must present a consistent design, functionality, and user experience across all components. Team collaboration is essential to ensure visual and functional coherence.

## Feedback and Progress

Each team member is expected to:

- Actively seek and provide feedback on peers' modules during weekly studio sessions.
- Make regular, visible contributions to the project's development from Week 1 until submission.
- Support the overall team effort through ongoing communication, code reviews, and shared problem-solving.

Consistent engagement, peer feedback, and collaboration are key to achieving both individual and team success in this assessment.

## C. GENERAL, TECHNICAL AND FUNCTIONAL REQUIREMENTS

It is up to the team to decide on the specific theme, goals and approaches for your site. Examples could include:

- Project proposal platform: a crowdfunding platform where creators present project ideas and ask the public to financially support them.
- E-commerce Store Online: A platform for custom and diverse fashion products
- Virtual Learning Platform: An online education site supporting course registration, video lessons, and quizzes.

- Event Planning and Ticketing Service: A platform for organizing, promoting, and selling event tickets
- Environmental Data Visualization: A site that presents environmental data through interactive visualizations and community discussions.

Your project may also explore any original idea your team wishes to pursue, as long as it enables you to meet the full set of technical and functional requirements described in this assessment brief.

While your team has the freedom to decide on the overall concept and theme of your project, your chosen idea must incorporate all requirements outlined in this document.

It is not acceptable to exclude a required feature on the basis that it does not naturally fit your concept. For instance, the statement “Our idea doesn’t need a shopping cart” is not a valid justification for omitting that component.

If certain features do not align perfectly with your theme, you are expected to adapt or extend your idea so that it accommodates all assessment requirements.

This section is long and includes MANY requirements, as is fitting for a major assignment. You are strongly encouraged to read through them all, and then to refer to them **FREQUENTLY** over the course of the semester while working on your website.

## GENERAL REQUIREMENTS

- All submitted code **MUST** be your own work. Use of AI or AI assisted technologies for producing code or documents for your assignment is not allowed without explicit acknowledgement.
- You must **NOT** make use of external frameworks such as React, jQuery, or similar (unless specifically in this specification or in the syllabus of the course).
- Your final website **MUST** be hosted and run on a server during the presentation.
- Any backend data **MUST** be stored using a MongoDB Atlas database.
- Images and files should be stored neatly in a relevant directory in your project, but the links (URLs) of those images and files are stored in the database.

## TECHNICAL REQUIREMENTS

- Server-side components developed using NodeJS.
- Client-side components developed using HTML5, CSS, and JavaScript.
- Database components developed using MongoDB Atlas.
- You are **NOT PERMITTED** to use technologies that are not taught in the course

## FUNCTIONAL REQUIREMENTS FOR ALL WEBSITES

- Professional design
  - Appropriate and appealing color scheme, fonts and layout.
  - Consistent application of accessibility considerations and requirements.

## MODULES (For Individual)

- The project consists of **individual modules** and **one shared module**.
- **Each team member** must be responsible for **a different module**.
- You **cannot duplicate** a module type (e.g., two discussion forums are not allowed).
- The descriptions provided below include details related to the **final project features** (e.g., “add,” “reply,” etc.) to help you understand the overall goal and context.

## Module 1: Discussion forum

Your website must include a Discussion Forum that allows users to collaboratively post and reply to one another through threaded discussions.

### *Main data about a post*

Each post, whether it begins a new thread or serves as a reply, must include:

- A title
- Post content
- Image content
- The author of the post
- A timestamp indicating when the post was made

### *Required Functionality*

The forum must allow users to:

- Create a new discussion thread with a subject and an initial post
- Reply to an existing thread or post
- Edit/Update all main data of their own posts or threads
- Delete their own posts (removed from public view but retained in the database for auditing purposes)
- Sort and filter threads based on:
  - Thread title
  - Text content within posts
  - Date of the most recent post
  - Date of the oldest post

### *Data Management*

All discussion data must be securely stored and managed in a MongoDB database.

## Module 2: Shopping cart

Your website must allow users to browse products, add them to a shopping cart, and simulate a checkout process.

### *Main data about an item in a shopping cart*

Each item in the cart must include:

- Text details (e.g., color, size, etc.)
- Product image
- Price
- Quantity

### *Required Functionality*

The shopping cart must allow users to:

- Create a new cart by adding an item
- Retrieve a list of available products to add to the cart
- Update all main data of a cart item (excluding product details such as the image, which remain fixed)

- Delete items from the cart
- Sort or filter items in the cart by:
  - Product title
  - Quantity
  - Price
- Simulate a purchase by submitting:
  - Credit card details
  - Delivery information

Upon submission, display a confirmation page summarizing the completed order.

#### *Data Management*

All product, cart, and order data must be securely stored and managed in a MongoDB database.

### **Module 3: Blog**

Your website must include a Blogging Platform that allows users to create, manage, and share personal or public blog posts.

#### *Main data about a blog*

Each blog post must include:

- Title
- Author
- Date added
- Tags or categories
- Main text content
- Image content

#### *Required Functionality*

The blogging platform must allow users to:

- Create a new blog post including all main components listed above
- View blog posts in two modes:
  - A list view displaying post previews (title, summary, thumbnail, and date)
  - A detailed view displaying the full content and comments
- Edit their own posts
- Delete their own posts
- Filter or search blog posts based on any of the main components

#### *Data Management*

All blog data must be securely stored and managed in a MongoDB database.

### **Module 4: Product review and rating**

Your website must include a Review and Rating System that allows users to submit feedback and star ratings for products they have purchased or used.

#### *Main data about a review*

Each review must include:

- Title
- Description
- Star rating
- Reviewer name
- Date added
- Image content

#### *Required Functionality*

The review system must allow users to:

- Create a new review including all main components listed above
- View reviews in two modes:
  - A list view showing review previews (title, summary, thumbnail, and date)
  - A detailed view showing the full review content
- Edit their own reviews
- Delete their own reviews
- Filter or search reviews by any of the main components

#### *Data Management*

All review data must be securely stored and managed in a MongoDB database.

## **Module 5: Wishlist**

The wishlist module adds, removes, and displays products which the users are interested in and therefore want to save somewhere in the application for later.

#### *Main data about an item in the wishlist*

- Name and image of the item
- Reference to the original item in the application that the users can access the item's details
- Statistics for wishlist items, e.g., the number of users who have added a particular item to their wishlist, the number of times a particular item has been added to shopping cart or bought.

#### *Required Functionality*

- Add a product to the wishlist.
- Remove a product from the wishlist.
- View all wishlist items.
- Prevent duplicate wishlist entries for the same user and product.
- Move a wishlist item to the shopping cart.
- Mark items as purchased.

## SHARED MODULES (For team collective development)

The following modules are shared and collectively developed by the whole team

### Shared module: user account management

Your website must include a Shared User Account System that manages authentication, access control, and user profile information across all modules.

#### *Required Functionality*

The system must allow users to:

- Register a new account with the following details:
  - Unique username
  - Securely hashed password
  - Valid email address
  - Short description
- Log in using valid credentials
- Log out securely from any session
- Reset password if forgotten
- View and edit their own profile information, including:
  - Name
  - Description
  - Email address
  - Profile picture
  - Password
- Delete or deactivate their own account, with appropriate warnings and confirmation prompts

#### *System Requirements*

The user account system must enforce:

- Unique usernames and valid email formats
- Secure password storage through hashing
- Session management with proper access control and session termination
- Role-based access control where applicable (e.g., admin vs. standard user)

#### *Data Management and Security*

All user account data must be securely stored and managed in a MongoDB database.

### Shared module: Administration

Your site must include a module for a site administrator to manage all aspects of the website. This must include the abilities to:

- Lock and unlock user accounts. The user locked cannot perform any operations in the application.

### Shared module: Sitemap

A sitemap is a page or file that lists all important sections of a website. It helps both users and search engines understand the site's structure, hierarchy, and navigation paths.

Your website must include an automatically generated HTML sitemap that provides users with an overview of all key pages and content across the site.

#### *Requirements:*

The sitemap page is expected to:

- Include all main navigation links, thread links, and post links
- Provide a clear, visual, and clickable overview of the website's structure
- Be automatically generated, ensuring new pages are added dynamically without requiring manual updates or hard-coded links
- List all public and static key pages to serve as an alternative navigation tool for users

## ADVANCED REQUIREMENTS

Your team will propose and implement one or more advanced features that align with the overall theme and purpose of your website.

As noted in the introduction, it is up to your team to agree on the overall purpose, aims and goals of your website.

While the previously listed requirements define the minimum functionality that every website must include, this component provides an opportunity to extend your project with innovative and higher-level features that demonstrate creativity and technical depth.

### **Expectations**

Your advanced feature(s) should:

- Implement aspects of web technology, design, or interaction that go beyond the core requirements listed earlier
- Substantially enhance or extend one or more of the minimum requirements to improve usability, functionality, or user experience
- Implement a specific algorithm (e.g., searching, matching, etc.) that integrates data within the application or with other systems to provide insights for users

## D. ASSESSMENT COMPONENTS

The project will involve three scaffolded components, corresponding to three assessment tasks 1, 2, 3

1. Static website
2. Prototype: Dynamic front-end application
3. Presentation and demonstration

Before submitting, carefully review the assignment specification to ensure all requirements have been met.

### **Hosting Requirements**

- The final website must be hosted on a live server.
- It must remain accessible until the end of the semester and marking period.
- No updates are allowed after the submission deadline.

## E. GENERAL SUBMISSION INFORMATION FOR ALL ASSESSMENTS



Full submission instructions for each assessment will be provided under the “Assignments” section of the course Canvas. The instructions found on the course Canvas are the official and final assessment instructions, and any updates there supersede any submission instructions given in this document.

Carefully note each submission time (as given in the “Assignments” section of the course Canvas). If your work is submitted after the deadline, a late penalty of 25% of the total available marks will be deducted for each 24-hour period after the submission deadline. The advertised submission time will be rigorously applied. Being late by any amount of time will mean that at least one 24-hour period of mark penalties will apply.

Therefore, you are encouraged to plan your submission carefully, including allowing extra time for each group member to VERIFY that you have submitted correctly (e.g. by re-downloading your submitted files and checking that they are the correct versions). If you discover a problem (e.g. that you have submitted the wrong version or missed submitting some of the required files) after the assignment submission deadline, then late penalties will apply, as described above.

If unexpected circumstances beyond your control have affected your ability to complete an assessment, you may be eligible to apply for special consideration. Please carefully read the information about special consideration, including circumstances that may warrant special consideration, and the supporting documentation that is required; the page also includes the online special consideration application form and full instructions: <https://www.rmit.edu.vn/students/my-studies/assessment-and-exams/assessment>

## **F. ACADEMIC INTEGRITY AND PLAGIARISM (STANDARD WARNING)**

Academic integrity is about honest presentation of your academic work. It means acknowledging the work of others while developing your own insights, knowledge and ideas. You should take extreme care that you have:

- Acknowledged words, data, diagrams, models, frameworks and/or ideas of others you have quoted (i.e. directly copied), summarized, paraphrased, discussed or mentioned in your assessment through the appropriate referencing methods,
- Provide a reference list of the publication details so your reader can locate the source if necessary. This includes material taken from Internet sites.

If you do not acknowledge the sources of your material, you may be accused of plagiarism because you have passed off the work and ideas of another person without appropriate referencing, as if they were your own.

RMIT University treats plagiarism as a very serious offense constituting misconduct. Plagiarism covers a variety of inappropriate behaviors, including:

- Failure to properly document a source
- Copyright material from the internet or databases
- Collusion between students

For further information on academic integrity policies and procedures, please refer to <https://www.rmit.edu.vn/students/my-studies/assessment-and-exams/assessment>

– The end of the Assessment Requirements –